

SQL Exercise 3Case Statements

Q1:

```
SELECT product_name,  
       price ,  
       CASE  
         WHEN price > 1000 THEN 'Expensive'  
         WHEN price BETWEEN 100 AND 1000 THEN  
           'Mid-range'  
         WHEN price < 100 THEN 'Budget'  
       END AS price-category  
FROM products ;
```

R: products

Product_name	Price	Price-category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

Q2:

```
SELECT customer_name,  
       amount ,  
       CASE  
         WHEN amount >= 1000 THEN 'High Value'  
         WHEN amount BETWEEN 500 - 999.99  
           THEN 'Medium Value'  
         WHEN amount < 500 THEN 'Low Value'
```


END AS order-value-category
FROM orders;

R: orders

customer_name	amount	order-value-category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

Q3:

```

SELECT emp_name,
       department,
       salary,
       CASE
         WHEN department = 'IT' AND salary > 80000
         THEN 'Senior IT'
         WHEN department = 'HR' AND salary > 55000
         THEN 'Experienced HR'
         ELSE 'Staff'
       END AS position-level
FROM employees;
```

R: employees

emp_name	department	salary	position-level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Q4:

```
SELECT student_name,  
       score,  
       CASE  
         WHEN score >= 90 THEN 'A'  
         WHEN score BETWEEN 80 AND 89 THEN 'B'  
         WHEN score BETWEEN 70 AND 79 THEN 'C'  
         WHEN score BETWEEN 60 AND 69 THEN 'D'  
         WHEN score < 60 THEN 'F'  
       END AS grade  
FROM students;
```

R: students

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Q5:

```
SELECT *,  
       CASE  
         WHEN delivery_time_minutes <= 30 THEN 'Fast'  
         WHEN delivery_time_minutes BETWEEN 31 AND 60  
           THEN 'On Time'  
         WHEN delivery_time_minutes > 60 THEN 'Late'  
       END AS performance  
FROM deliveries;
```


R: deliveries

delivery-id	delivery-time-minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Q6:

```
SELECT issue_type,  
       priority,  
       CASE
```

```
         WHEN priority = 3 THEN 'High'
```

```
         WHEN priority = 2 THEN 'Medium'
```

```
         WHEN priority = 1 THEN 'Low'
```

```
       END AS priority-label
```

```
FROM tickets;
```

R: tickets

issue_type	priority	priority-label
Login issue	1	Low
Server down	3	High
slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

Q7:

~~SELECT * FROM tickets,~~

~~days-present * 100 / (number-days) = AB~~

Q7:

```
SELECT student_id,  
       days_present * 100.0 / total_days  
       AS attendance_percentage,  
       CASE  
         WHEN days_present * 100.0 / total_days >= 90 THEN 'Excellent'  
         WHEN days_present * 100.0 / total_days >= 75 THEN 'Good'  
         WHEN days_present * 100.0 / total_days < 75 THEN 'Needs Improvement'  
       END AS attendance_status  
FROM attendance;
```

R: attendance

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Q8:

```
SELECT *,  
       CASE  
         WHEN stock_qty = 0 THEN 'Out of Stock'  
         WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'  
         WHEN stock_qty > 5 THEN 'In Stock'  
       END AS products_inventory stock_status  
FROM products_inventory;
```


R: products inventory

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Q9:

```
SELECT subject,  
        enrolled_students,  
        CASE  
            WHEN num_of_students >= 25 THEN 'Large'  
            WHEN num_of_students BETWEEN 10 AND 24  
            THEN 'Medium'  
            WHEN num_of_students < 10 THEN 'Small'  
        END AS class_size_category  
FROM classes;
```

R: classes

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Q10:

```
SELECT payment_id,  
       payment_method,  
       amount,  
       CASE  
         WHEN payment_method = 'cash' AND amount >=  
           200 THEN 'Eligible for Discount'  
         ELSE 'Not Eligible'  
       END AS discount_eligibility  
FROM payments;
```

R: payments

payment_id	payment_method	amount	discount_eligibility
1	Card	50	Not Eligible
2	Cash	200	Eligible for Discount
3	Card	150	Not Eligible
4	PayPal	75	Not Eligible
5	Cash	300	Eligible for Discount